Hybrid Air Purification & Direct Air Capture Tower

SYSTEM TELEMETRY & TECHNO-ECONOMIC PROTOTYPE SPECIFICATION

Unit: HT-URBAN-042 | OPERATIONAL

EST. CAPTURE COST
\$102 / tonne
-80% vs. \$500-\$600 Standalone DAC

AIR FILTRATION EFF.
94.8%
PM2.5 Reduced from 85 to 12 µg/m³

CRADLE-TO-GRAVE DAC
92.1% EFF.
43.2 kg CO_2 / 24h (1.8 kg/hr avg)

STREAM CO_2 PURITY
99.2%
Amino Acid + Polymer Contactor

Shared Infrastructure & Energy Integration

Airflow Volume & Blower Speed	1,200 m³/hr @ 1,850 RPM
Waste Heat Thermal Recovery	72°C Source (88.4% Eff)
Sorbent Regeneration Temp	68°C Integrated Low-Temp
CapEx & Fan Ducting Savings	-45% vs Dedicated Infrastructure

Techno-Economic Benchmark

Metric	Standalone DAC	Hybrid Tower (HT-042)
Cost / Tonne	\$500 – \$600	\$102 / tonne
Energy Source	Dedicated Plant	HVAC / Geothermal Heat
Land Footprint	Large Plot	Compact Vertical Column
Health Co-Benefits	None (Pure CO_2)	PM2.5, PM10 & NOx Filtering

Modular Urban Deployment
Plaza & Transit Hub Co-location with HVAC Systems
+\$29.00/t Net Profit

Downstream CO_2 Offtake & Commercial Value

Concrete Mineralization
Increases compressive strength by 15%

Synthetic E-Fuels
Green hydrogen mix for net-zero aviation

Geological Storage
Deep basalt trapping under 45Q (\$180/t)

Global Scale Goal: 596 Mt CO_2/year potential across 10,000+ urban units

Image Sources

https://thejewelerblog.wordpress.com/wp-content/uploads/2015/09/tower2.jpg

Source: thejewelerblog.wordpress.com